

(i) Advantages of AAC

1. The most important advantage is its lower environmental impact.
2. Improved thermal efficiency reduces the heating and cooling load in buildings.
3. Excellent soundproofing material and acoustic insulator.
4. Highly fire and termite-resistant.
5. Manufactured in a variety of forms and sizes.
6. Reduces project cost.
7. Absorbs moisture from air and releases humidity.
8. Instalation of electrical and plumbing lines are easy.
9. Shipping and handling is more economical.
10. It is a durable material.

2.2.3 Fibre Reinforced Concrete (FRC)

The inherent weakness of concrete is development of micro-cracks due to drying shrinkage. The technique of using fibres in the mix help in overcoming this weakness to a great extent. Fibre Reinforced Concrete is a composite material consisting of cement mortar or concrete with uniformly distributed discrete fibres of high tensile strength.



The various types of fibres used in FRC are steel, Polypropylene, Nylon, Asbestos, Coir, Glass, Carbon, etc.

(i) Advantages of FRC

1. It increases the tensile strength of the concrete.
2. It reduce the voids in concrete.
3. It increases the durability of the concrete.
4. It has been recognized that the addition of small, closely spaced and uniformly dispersed fibres to concrete would act as crack arrester.
5. Improve its static and dynamic properties.

2.2.4 Self Compacting Concrete (SCC)

Self compacting concrete is highly flowable, non-segregating concrete that can spread into place, fill form work and encapsulate the reinforcement without any mechanical consolidation. The spread of self compacting concrete typically ranges from 1 to 32 inches depending on the requirements for the construction. Materials used for self compacting concrete are cement (43 or 53 grade), aggregates (less than 20 mm), Water and super plasticizers (chemical admixtures).

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